

# Nonparametric\_estimation\_survival\_function

## Advanced Health Data Analysis: Survival Analysis, Course 2025/26

### 23.2.2026: Nonparametric estimation of the survival function

```
load("Ahda_RLab1.RData")
ls()
```

```
[1] "remission"
```

#### The data set

```
str(remission)
```

```
'data.frame':  42 obs. of  5 variables:
 $ pair  : num  1 2 3 4 5 6 7 8 9 10 ...
 ..- attr(*, "label")= chr "Pair ID"
 $ time  : num  1 22 3 12 8 17 2 11 8 12 ...
 ..- attr(*, "label")= chr "Time until relapse"
 ..- attr(*, "units")= chr "Week"
 $ cens  : num  1 1 1 1 1 1 1 1 1 1 ...
 ..- attr(*, "label")= chr "Relapse indicator"
 ..- attr(*, "levels")= chr [1:2] "No relapse" "Relapse"
 $ treat : Factor w/ 2 levels "Control","6-MP": 1 1 1 1 1 1 1 1 1 1 ...
 ..- attr(*, "label")= chr "Treatment"
 $ status: num  1 2 2 2 2 1 2 2 2 2 ...
 ..- attr(*, "label")= chr "Disease stage"
 ..- attr(*, "levels")= chr [1:2] "Partial remission" "Complete remission"
 - attr(*, "comment")= chr "Data of a clinical trial on leukemia"
```

```
Hmisc::Label(remission)
```

```
label(pair) <- 'Pair ID'
label(time) <- 'Time until relapse'
label(cens) <- 'Relapse indicator'
label(treat)  <- 'Treatment'
label(status) <- 'Disease stage'
```

```
head(remission)
```

```
  pair time cens  treat status
1    1   1    1  Control     1
2    2  22    1  Control     2
3    3   3    1  Control     2
4    4  12    1  Control     2
5    5   8    1  Control     2
6    6  17    1  Control     1
```

```
summary(remission)
```

| pair       | time          | cens            | treat      | status         |
|------------|---------------|-----------------|------------|----------------|
| Min. : 1   | Min. : 1.00   | Min. : 0.0000   | Control:21 | Min. : 1.000   |
| 1st Qu.: 6 | 1st Qu.: 6.00 | 1st Qu.: 0.0000 | 6-MP :21   | 1st Qu.: 2.000 |
| Median :11 | Median :10.50 | Median : 1.0000 |            | Median : 2.000 |
| Mean :11   | Mean :12.88   | Mean : 0.7143   |            | Mean : 1.762   |
| 3rd Qu.:16 | 3rd Qu.:18.50 | 3rd Qu.: 1.0000 |            | 3rd Qu.: 2.000 |
| Max. :21   | Max. :35.00   | Max. : 1.0000   |            | Max. : 2.000   |

## Quick data exploration

### Number of relapses per group

```
with(remission, table(treat, cens))
```

```
      cens
treat    0  1
Control  0 21
6-MP    12  9
```

makes us assume possibility 1

### Mean follow-up time per group

```
with(remission, tapply(time, treat, summary))
```

```
$Control
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1.000  4.000   8.000  8.667 12.000 23.000

$`6-MP`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 6.0    9.0   16.0   17.1   23.0   35.0
```

Mean follow-up time for 6MP is larger and looks like the treatment was beneficial but we need a closer look at censored events.

## Kaplan-Meier estimation of $S(t)$ with R

```
library(survival)
```

### The Surv object

```
with(remission, Surv(time, cens))
```

```
[1] 1 22 3 12 8 17 2 11 8 12 2 5 4 15 8 23 5 11 4
[20] 1 8 10 7 32+ 23 22 6 16 34+ 32+ 25+ 11+ 20+ 19+ 6 17+ 35+ 6
[39] 13 9+ 6+ 10+
```

- indicates right censored observation

```
srem <- with(remission, Surv(time, cens))
srem
```

```
[1] 1 22 3 12 8 17 2 11 8 12 2 5 4 15 8 23 5 11 4
[20] 1 8 10 7 32+ 23 22 6 16 34+ 32+ 25+ 11+ 20+ 19+ 6 17+ 35+ 6
[39] 13 9+ 6+ 10+
```

```
str(srem)
```

```
'Surv' num [1:42, 1:2] 1 22 3 12 8 17 2 11 8 12 ...
- attr(*, "dimnames")=List of 2
..$ : NULL
..$ : chr [1:2] "time" "status"
- attr(*, "type")= chr "right"
```

```

- attr(*, "inputAttributes")=List of 2
..$ time :List of 2
.. ..$ label: chr "Time until relapse"
.. ..$ units: chr "Week"
..$ event:List of 2
.. ..$ label : chr "Relapse indicator"
.. ..$ levels: chr [1:2] "No relapse" "Relapse"

```

## The Kaplan-Meier estimate is obtained with the function survfit()

```
survfit(srem ~ 1) #~1 to use whole sample
```

Call: survfit(formula = srem ~ 1)

```

      n events median 0.95LCL 0.95UCL
[1,] 42      30      12       8      22

```

- 42 samples, 30 died, 42-30=12 are right censored
- median survival time with 95% interval
- see estimation of median

## More information is shown with the function summary.survfit()

```
svf <- survfit(srem ~ 1)
summary(svf)
```

Call: survfit(formula = srem ~ 1)

| time | n.risk | n.event | survival | std.err | lower | 95% CI | upper | 95% CI |
|------|--------|---------|----------|---------|-------|--------|-------|--------|
| 1    | 42     | 2       | 0.952    | 0.0329  |       | 0.8901 |       | 1.000  |
| 2    | 40     | 2       | 0.905    | 0.0453  |       | 0.8202 |       | 0.998  |
| 3    | 38     | 1       | 0.881    | 0.0500  |       | 0.7883 |       | 0.985  |
| 4    | 37     | 2       | 0.833    | 0.0575  |       | 0.7279 |       | 0.954  |
| 5    | 35     | 2       | 0.786    | 0.0633  |       | 0.6709 |       | 0.920  |
| 6    | 33     | 3       | 0.714    | 0.0697  |       | 0.5899 |       | 0.865  |
| 7    | 29     | 1       | 0.690    | 0.0715  |       | 0.5628 |       | 0.845  |
| 8    | 28     | 4       | 0.591    | 0.0764  |       | 0.4588 |       | 0.762  |
| 10   | 23     | 1       | 0.565    | 0.0773  |       | 0.4325 |       | 0.739  |
| 11   | 21     | 2       | 0.512    | 0.0788  |       | 0.3783 |       | 0.692  |
| 12   | 18     | 2       | 0.455    | 0.0796  |       | 0.3227 |       | 0.641  |
| 13   | 16     | 1       | 0.426    | 0.0795  |       | 0.2958 |       | 0.615  |
| 15   | 15     | 1       | 0.398    | 0.0791  |       | 0.2694 |       | 0.588  |
| 16   | 14     | 1       | 0.369    | 0.0784  |       | 0.2437 |       | 0.560  |
| 17   | 13     | 1       | 0.341    | 0.0774  |       | 0.2186 |       | 0.532  |
| 22   | 9      | 2       | 0.265    | 0.0765  |       | 0.1507 |       | 0.467  |
| 23   | 7      | 2       | 0.189    | 0.0710  |       | 0.0909 |       | 0.395  |

- time = uncensored event times
- n risk = people at risk
- point estimation of survival probability at that time point
- first censored observation between week 6 and 7
- after week 23: still 5 censored observations left
- function stays constant up to week 35 (times of censored observations)

## Some options of the function summary.survfit():

```
summary(svf, censored=TRUE)
```

Call: survfit(formula = srem ~ 1)

| time | n.risk | n.event | survival | std.err | lower | 95% CI | upper | 95% CI |
|------|--------|---------|----------|---------|-------|--------|-------|--------|
| 1    | 42     | 2       | 0.952    | 0.0329  |       | 0.8901 |       | 1.000  |
| 2    | 40     | 2       | 0.905    | 0.0453  |       | 0.8202 |       | 0.998  |

|    |    |   |       |        |        |       |
|----|----|---|-------|--------|--------|-------|
| 3  | 38 | 1 | 0.881 | 0.0500 | 0.7883 | 0.985 |
| 4  | 37 | 2 | 0.833 | 0.0575 | 0.7279 | 0.954 |
| 5  | 35 | 2 | 0.786 | 0.0633 | 0.6709 | 0.920 |
| 6  | 33 | 3 | 0.714 | 0.0697 | 0.5899 | 0.865 |
| 7  | 29 | 1 | 0.690 | 0.0715 | 0.5628 | 0.845 |
| 8  | 28 | 4 | 0.591 | 0.0764 | 0.4588 | 0.762 |
| 9  | 24 | 0 | 0.591 | 0.0764 | 0.4588 | 0.762 |
| 10 | 23 | 1 | 0.565 | 0.0773 | 0.4325 | 0.739 |
| 11 | 21 | 2 | 0.512 | 0.0788 | 0.3783 | 0.692 |
| 12 | 18 | 2 | 0.455 | 0.0796 | 0.3227 | 0.641 |
| 13 | 16 | 1 | 0.426 | 0.0795 | 0.2958 | 0.615 |
| 15 | 15 | 1 | 0.398 | 0.0791 | 0.2694 | 0.588 |
| 16 | 14 | 1 | 0.369 | 0.0784 | 0.2437 | 0.560 |
| 17 | 13 | 1 | 0.341 | 0.0774 | 0.2186 | 0.532 |
| 19 | 11 | 0 | 0.341 | 0.0774 | 0.2186 | 0.532 |
| 20 | 10 | 0 | 0.341 | 0.0774 | 0.2186 | 0.532 |
| 22 | 9  | 2 | 0.265 | 0.0765 | 0.1507 | 0.467 |
| 23 | 7  | 2 | 0.189 | 0.0710 | 0.0909 | 0.395 |
| 25 | 5  | 0 | 0.189 | 0.0710 | 0.0909 | 0.395 |
| 32 | 4  | 0 | 0.189 | 0.0710 | 0.0909 | 0.395 |
| 34 | 2  | 0 | 0.189 | 0.0710 | 0.0909 | 0.395 |
| 35 | 1  | 0 | 0.189 | 0.0710 | 0.0909 | 0.395 |

- from week 23 on: no events but censored observations
- function is not defined beyond week 35

when I am only interested in the survival probability monthly:

```
summary(svf, times = seq(4, 24, 4)) #range, steps
```

Call: survfit(formula = srem ~ 1)

| time | n.risk | n.event | survival | std.err | lower | 95% CI | upper | 95% CI |
|------|--------|---------|----------|---------|-------|--------|-------|--------|
| 4    | 37     | 7       | 0.833    | 0.0575  |       | 0.7279 |       | 0.954  |
| 8    | 28     | 10      | 0.591    | 0.0764  |       | 0.4588 |       | 0.762  |
| 12   | 18     | 5       | 0.455    | 0.0796  |       | 0.3227 |       | 0.641  |
| 16   | 14     | 3       | 0.369    | 0.0784  |       | 0.2437 |       | 0.560  |
| 20   | 10     | 1       | 0.341    | 0.0774  |       | 0.2186 |       | 0.532  |
| 24   | 5      | 4       | 0.189    | 0.0710  |       | 0.0909 |       | 0.395  |

```
summary(svf, scale = 4)
```

Call: survfit(formula = srem ~ 1)

| time | n.risk | n.event | survival | std.err | lower | 95% CI | upper | 95% CI |
|------|--------|---------|----------|---------|-------|--------|-------|--------|
| 0.25 | 42     | 2       | 0.952    | 0.0329  |       | 0.8901 |       | 1.000  |
| 0.50 | 40     | 2       | 0.905    | 0.0453  |       | 0.8202 |       | 0.998  |
| 0.75 | 38     | 1       | 0.881    | 0.0500  |       | 0.7883 |       | 0.985  |
| 1.00 | 37     | 2       | 0.833    | 0.0575  |       | 0.7279 |       | 0.954  |
| 1.25 | 35     | 2       | 0.786    | 0.0633  |       | 0.6709 |       | 0.920  |
| 1.50 | 33     | 3       | 0.714    | 0.0697  |       | 0.5899 |       | 0.865  |
| 1.75 | 29     | 1       | 0.690    | 0.0715  |       | 0.5628 |       | 0.845  |
| 2.00 | 28     | 4       | 0.591    | 0.0764  |       | 0.4588 |       | 0.762  |
| 2.50 | 23     | 1       | 0.565    | 0.0773  |       | 0.4325 |       | 0.739  |
| 2.75 | 21     | 2       | 0.512    | 0.0788  |       | 0.3783 |       | 0.692  |
| 3.00 | 18     | 2       | 0.455    | 0.0796  |       | 0.3227 |       | 0.641  |
| 3.25 | 16     | 1       | 0.426    | 0.0795  |       | 0.2958 |       | 0.615  |
| 3.75 | 15     | 1       | 0.398    | 0.0791  |       | 0.2694 |       | 0.588  |
| 4.00 | 14     | 1       | 0.369    | 0.0784  |       | 0.2437 |       | 0.560  |
| 4.25 | 13     | 1       | 0.341    | 0.0774  |       | 0.2186 |       | 0.532  |
| 5.50 | 9      | 2       | 0.265    | 0.0765  |       | 0.1507 |       | 0.467  |
| 5.75 | 7      | 2       | 0.189    | 0.0710  |       | 0.0909 |       | 0.395  |

```
survfit(srem ~ 1)
```

Call: survfit(formula = srem ~ 1)

```
      n events median 0.95LCL 0.95UCL
[1,] 42      30      12       8       22
```

## Estimation of the restricted mean survival time (often not necessary for basic info)

```
print(svf, rmean = "common") #up to week 35
```

```
Call: survfit(formula = srem ~ 1)
```

```
      n events rmean* se(rmean) median 0.95LCL 0.95UCL
[1,] 42      30  15.3      1.86     12      8      22
* restricted mean with upper limit = 35
```

```
print(svf, rmean = 50) #extend to week 50 cuase it is still positive before, ASSUMING maximal survival
```

```
Call: survfit(formula = srem ~ 1)
```

```
      n events rmean* se(rmean) median 0.95LCL 0.95UCL
[1,] 42      30  18.2      2.81     12      8      22
* restricted mean with upper limit = 50
```

## Information saved in the survfit and summary.survfit objects

```
names(svf)
```

```
[1] "n"          "time"       "n.risk"     "n.event"    "n.censor"   "surv"
[7] "std.err"    "cumhaz"     "std.chaz"   "type"       "logse"      "conf.int"
[13] "conf.type"  "lower"      "upper"      "t0"         "call"
```

```
str(svf)
```

```
List of 17
 $ n          : int 42
 $ time       : num [1:24] 1 2 3 4 5 6 7 8 9 10 ...
 $ n.risk     : num [1:24] 42 40 38 37 35 33 29 28 24 23 ...
 $ n.event    : num [1:24] 2 2 1 2 2 3 1 4 0 1 ...
 $ n.censor   : num [1:24] 0 0 0 0 0 1 0 0 1 1 ...
 $ surv       : num [1:24] 0.952 0.905 0.881 0.833 0.786 ...
 $ std.err    : num [1:24] 0.0345 0.0501 0.0567 0.069 0.0806 ...
 $ cumhaz     : num [1:24] 0.0476 0.0976 0.1239 0.178 0.2351 ...
 $ std.chaz   : num [1:24] 0.0337 0.0488 0.0555 0.0674 0.0785 ...
 $ type       : chr "right"
 $ logse      : logi TRUE
 $ conf.int   : num 0.95
 $ conf.type  : chr "log"
 $ lower      : num [1:24] 0.89 0.82 0.788 0.728 0.671 ...
 $ upper      : num [1:24] 1 0.998 0.985 0.954 0.92 ...
 $ t0         : num 0
 $ call       : language survfit(formula = srem ~ 1)
 - attr(*, "class")= chr "survfit"
```

```
names(summary(svf))
```

```
[1] "n"          "time"       "n.risk"     "n.event"
[5] "n.censor"   "surv"       "std.err"    "cumhaz"
[9] "std.chaz"   "type"       "logse"      "conf.int"
[13] "conf.type"  "lower"      "upper"      "t0"
[17] "call"       "table"      "rmean.endtime"
```

```
str(summary(svf))
```

```
List of 19
 $ n          : int 42
 $ time       : num [1:17] 1 2 3 4 5 6 7 8 10 11 ...
 $ n.risk     : num [1:17] 42 40 38 37 35 33 29 28 23 21 ...
 $ n.event    : num [1:17] 2 2 1 2 2 3 1 4 1 2 ...
 $ n.censor   : num [1:17] 0 0 0 0 0 1 0 0 2 1 ...
 $ surv       : num [1:17] 0.952 0.905 0.881 0.833 0.786 ...
 $ std.err    : num [1:17] 0.0329 0.0453 0.05 0.0575 0.0633 ...
 $ cumhaz     : num [1:17] 0.0476 0.0976 0.1239 0.178 0.2351 ...
 $ std.chaz   : num [1:17] 0.0337 0.0488 0.0555 0.0674 0.0785 ...
 $ type       : chr "right"
 $ logse      : logi TRUE
```

```

$ conf.int      : num 0.95
$ conf.type     : chr "log"
$ lower        : num [1:17] 0.89 0.82 0.788 0.728 0.671 ...
$ upper        : num [1:17] 1 0.998 0.985 0.954 0.92 ...
$ t0           : num 0
$ call         : language survfit(formula = srem ~ 1)
$ table        : Named num [1:9] 42 42 42 30 15.3 ...
  .. attr(*, "names")= chr [1:9] "records" "n.max" "n.start" "events" ...
$ rmean.endtime: num 35
- attr(*, "class")= chr "summary.survfit"

```

## Treatment comparison with respect to S(t)

```
(svf2t <- survfit(srem ~ treat, remission))
```

Call: survfit(formula = srem ~ treat, data = remission)

|               | n  | events | median | 0.95LCL | 0.95UCL |
|---------------|----|--------|--------|---------|---------|
| treat=Control | 21 | 21     | 8      | 4       | 12      |
| treat=6-MP    | 21 | 9      | 23     | 16      | NA      |

```
summary(svf2t)
```

Call: survfit(formula = srem ~ treat, data = remission)

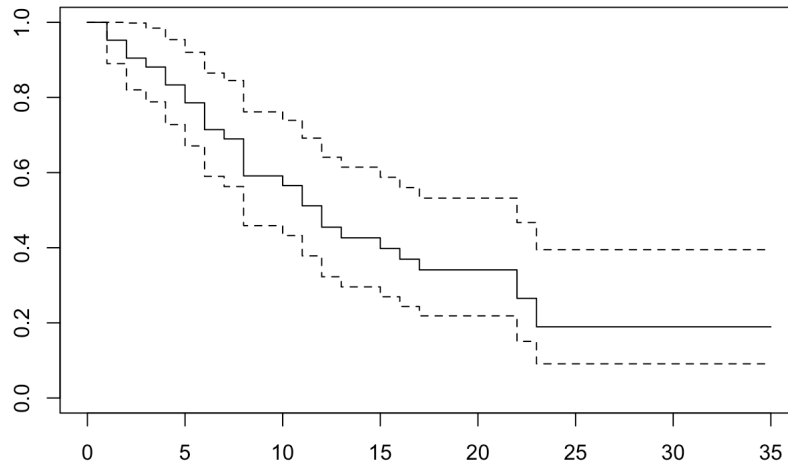
| treat=Control |        |         |          |         |              |              |  |
|---------------|--------|---------|----------|---------|--------------|--------------|--|
| time          | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 1             | 21     | 2       | 0.9048   | 0.0641  | 0.78754      | 1.000        |  |
| 2             | 19     | 2       | 0.8095   | 0.0857  | 0.65785      | 0.996        |  |
| 3             | 17     | 1       | 0.7619   | 0.0929  | 0.59988      | 0.968        |  |
| 4             | 16     | 2       | 0.6667   | 0.1029  | 0.49268      | 0.902        |  |
| 5             | 14     | 2       | 0.5714   | 0.1080  | 0.39455      | 0.828        |  |
| 8             | 12     | 4       | 0.3810   | 0.1060  | 0.22085      | 0.657        |  |
| 11            | 8      | 2       | 0.2857   | 0.0986  | 0.14529      | 0.562        |  |
| 12            | 6      | 2       | 0.1905   | 0.0857  | 0.07887      | 0.460        |  |
| 15            | 4      | 1       | 0.1429   | 0.0764  | 0.05011      | 0.407        |  |
| 17            | 3      | 1       | 0.0952   | 0.0641  | 0.02549      | 0.356        |  |
| 22            | 2      | 1       | 0.0476   | 0.0465  | 0.00703      | 0.322        |  |
| 23            | 1      | 1       | 0.0000   | NaN     | NA           | NA           |  |

| treat=6-MP |        |         |          |         |              |              |  |
|------------|--------|---------|----------|---------|--------------|--------------|--|
| time       | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 6          | 21     | 3       | 0.857    | 0.0764  | 0.720        | 1.000        |  |
| 7          | 17     | 1       | 0.807    | 0.0869  | 0.653        | 0.996        |  |
| 10         | 15     | 1       | 0.753    | 0.0963  | 0.586        | 0.968        |  |
| 13         | 12     | 1       | 0.690    | 0.1068  | 0.510        | 0.935        |  |
| 16         | 11     | 1       | 0.627    | 0.1141  | 0.439        | 0.896        |  |
| 22         | 7      | 1       | 0.538    | 0.1282  | 0.337        | 0.858        |  |
| 23         | 6      | 1       | 0.448    | 0.1346  | 0.249        | 0.807        |  |

- compared control and treatment group
- control: estimated survival function goes down to zero, all died
- No upper limit for median of treatment group cause the intervals in the bottom table do not drop below 0,5

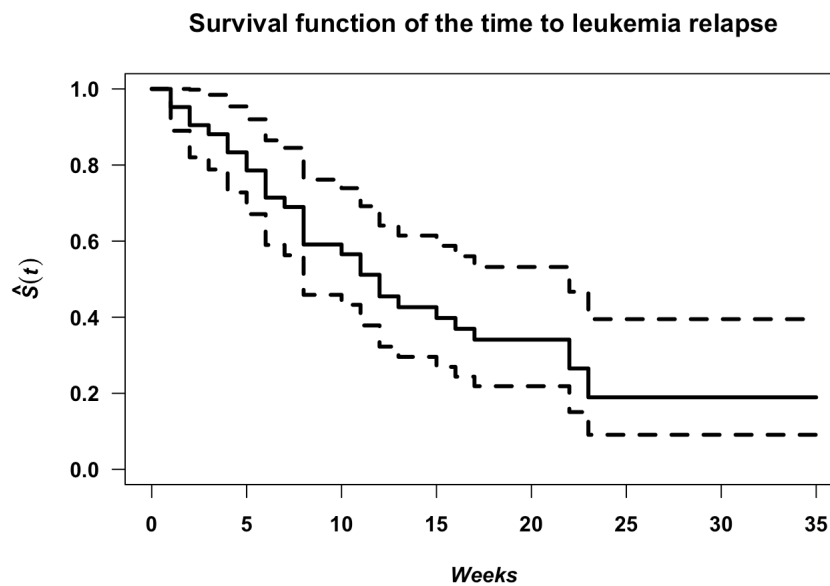
## Graphical representation of the estimated survival functions

```
plot(svf)
```



## A little bit nicer

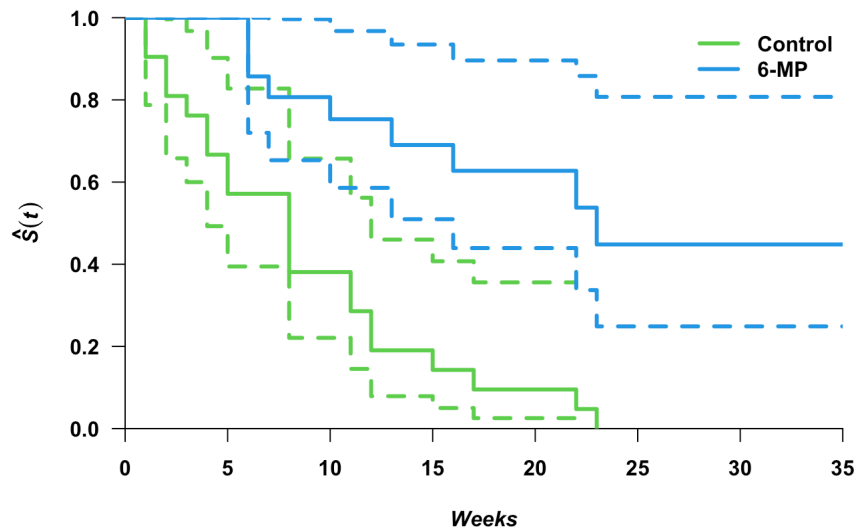
```
#windows(width = 10)
par(font = 2, font.axis = 2, font.lab = 4, las = 1, mar = c(5, 5, 4, 2))
plot(svf, lwd = 3, xlab = "Weeks", ylab = expression(bolditalic(hat(S)(t))))
title("Survival function of the time to leukemia relapse")
```



## Both treatment arms

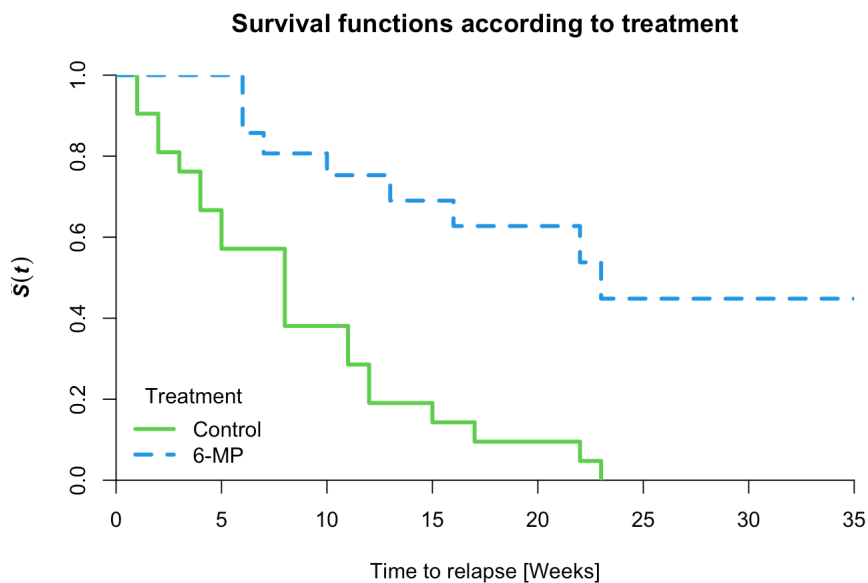
```
windows(width = 10)
```

```
par(font = 2, font.axis = 2, font.lab = 4, las = 1, mar = c(5, 5, 4, 2))
plot(svf2t, col = 3:4, conf.int = TRUE, lwd = 3, xlab = "Weeks", bty = "n",
     ylab = expression(bolditalic(hat(S)(t))), xaxs = "i", yaxs = "i")
legend("topright", levels(remission$treat), col = 3:4, lwd = 3, bty = "n")
```



### Maybe better without the confidence intervals

```
plot(svf2t, col = 3:4, xlab = "Time to relapse [Weeks]",
     ylab = expression(bolditalic(hat(S)(t))),
     lty = 1:2, lwd = 3, yaxs = "i", xaxs = "i", bty = "n")
title("Survival functions according to treatment")
legend("bottomleft", levels(remission$treat), title = "Treatment",
      bty = "n", col = 3:4, lty = 1:2, lwd = 3)
```



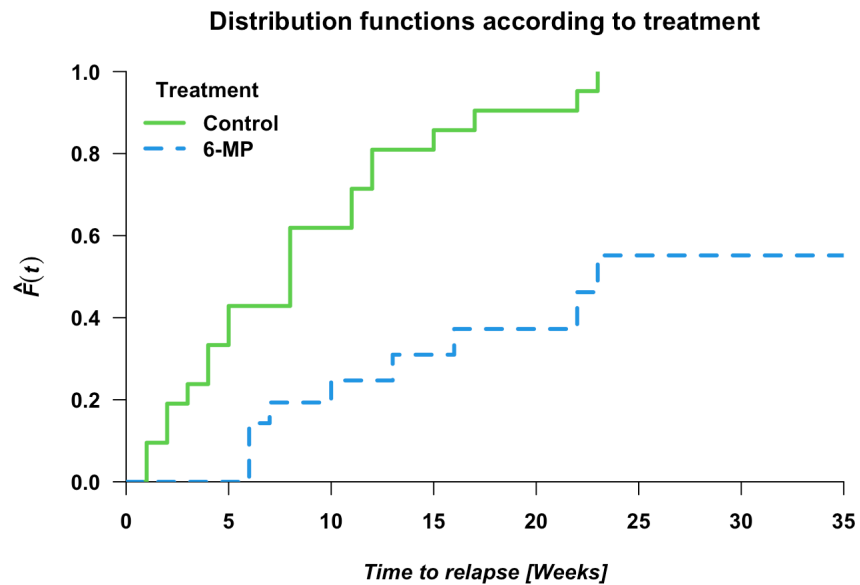
### Representation of the distribution function (cumulative incidence)

```
windows(width = 10)
```

```
par(font = 2, font.axis = 2, font.lab = 4, las = 1, mar = c(5, 5, 4, 2))
plot(svf2t, col = 3:4, xlab = "Time to relapse [Weeks]",
     ylab = expression(bolditalic(hat(S)(t))), yaxs = "i",
```



```
ylab = expression(bolditalic(hat(Lambda)(t))), las = 1,
lty = 1:2, fun = "event", lwd = 3, yaxs = "i", bty = "n") #define fun ="event"
title("Distribution functions according to treatment")
legend("topleft", levels(remission$treat), title = "Treatment",
      bty = "n", col = 3:4, lty = 1:2, lwd = 3)
```



## Representation of the cumulative hazard function

```
windows(width = 10) par(font = 2, font.axis = 2, font.lab = 4, las = 1, mar = c(5, 5, 4, 2)) plot(svf2t, col = 3:4, xlab =
"Time to relapse [Weeks]", ylab = expression(bolditalic(hat(Lambda)(t))), xaxs = "i", lty = 1:2, fun = ..., lwd = 3, yaxs = "i",
bty = "l") title("Cumulative hazards functions according to treatment") legend("topleft", levels(remission$treat), title =
"Treatment", bty = "n", col = 3:4, lty = 1:2, lwd = 3)
```